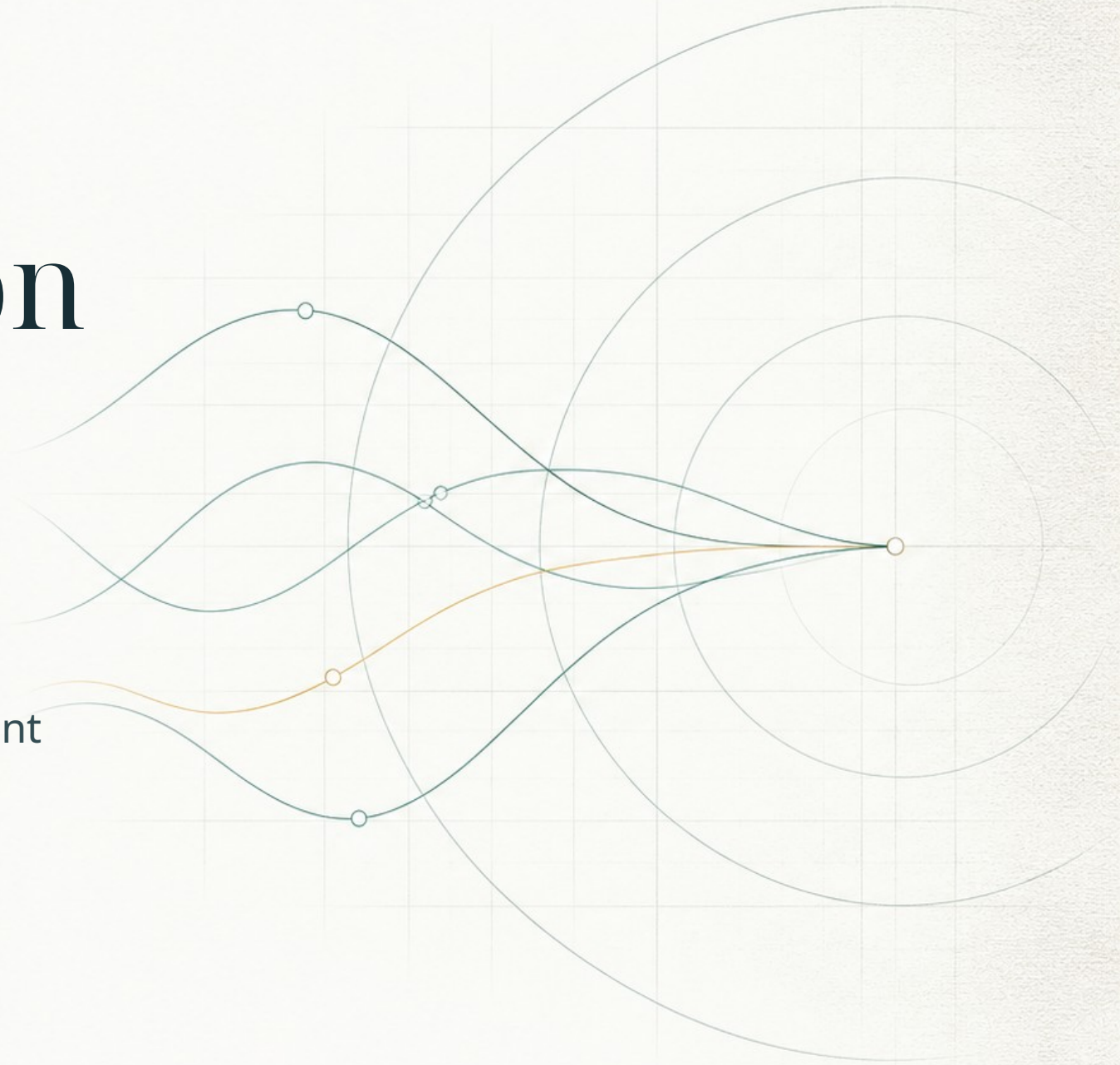
Payments:Unscripted panellist application

Can tokenization shrink the cash buffer?

Goods move in batches. Each verified payment can settle as it comes due.



Terence Heju

Founder, Connextium.xyz

The cash buffer depends on the wait

Treasury holds extra cash to cover uncertain outflows until incoming funds are usable.

$$SB = z\sigma\sqrt{T}$$

With the same safety factor and volatility, halving the wait reduces the buffer by 29%.

z **Safety factor**
How much forecast error the buffer should cover.

σ **Forecast volatility**
Standard deviation of net cash-flow forecast error per time step.

T **Funding window**
Time from invoice due time to spendable funds in this example.

The same square-root relationship appears in safety stock when replenishment lead time is fixed.

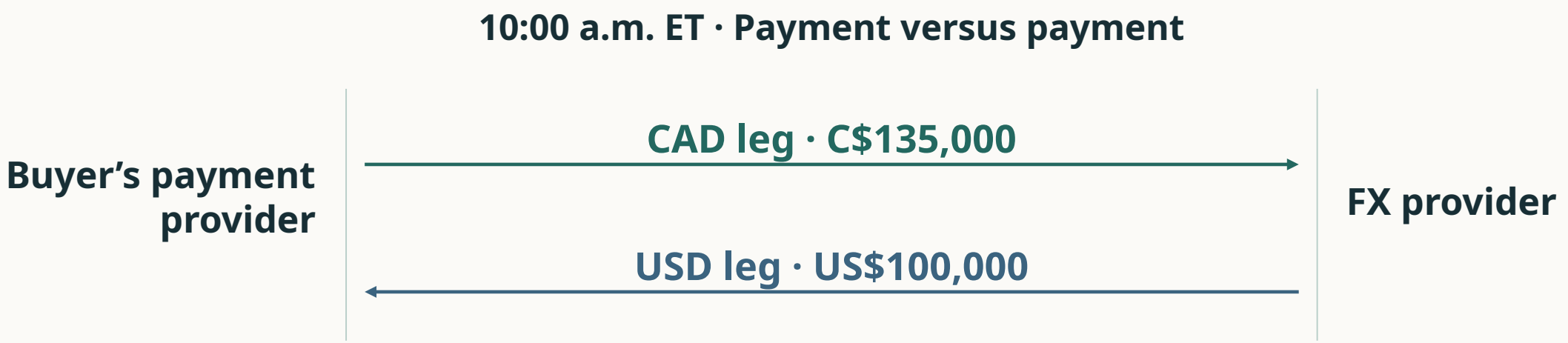
The model assumes independent forecast errors with constant variance per time step.

Whalen (1966) · Precautionary demand for cash

MIT · Safety stock

CAD and USD settle together

A Canadian buyer owes a U.S. supplier US$100,000 at 10:00 a.m. ET.



Each transfer becomes final only if the other does.

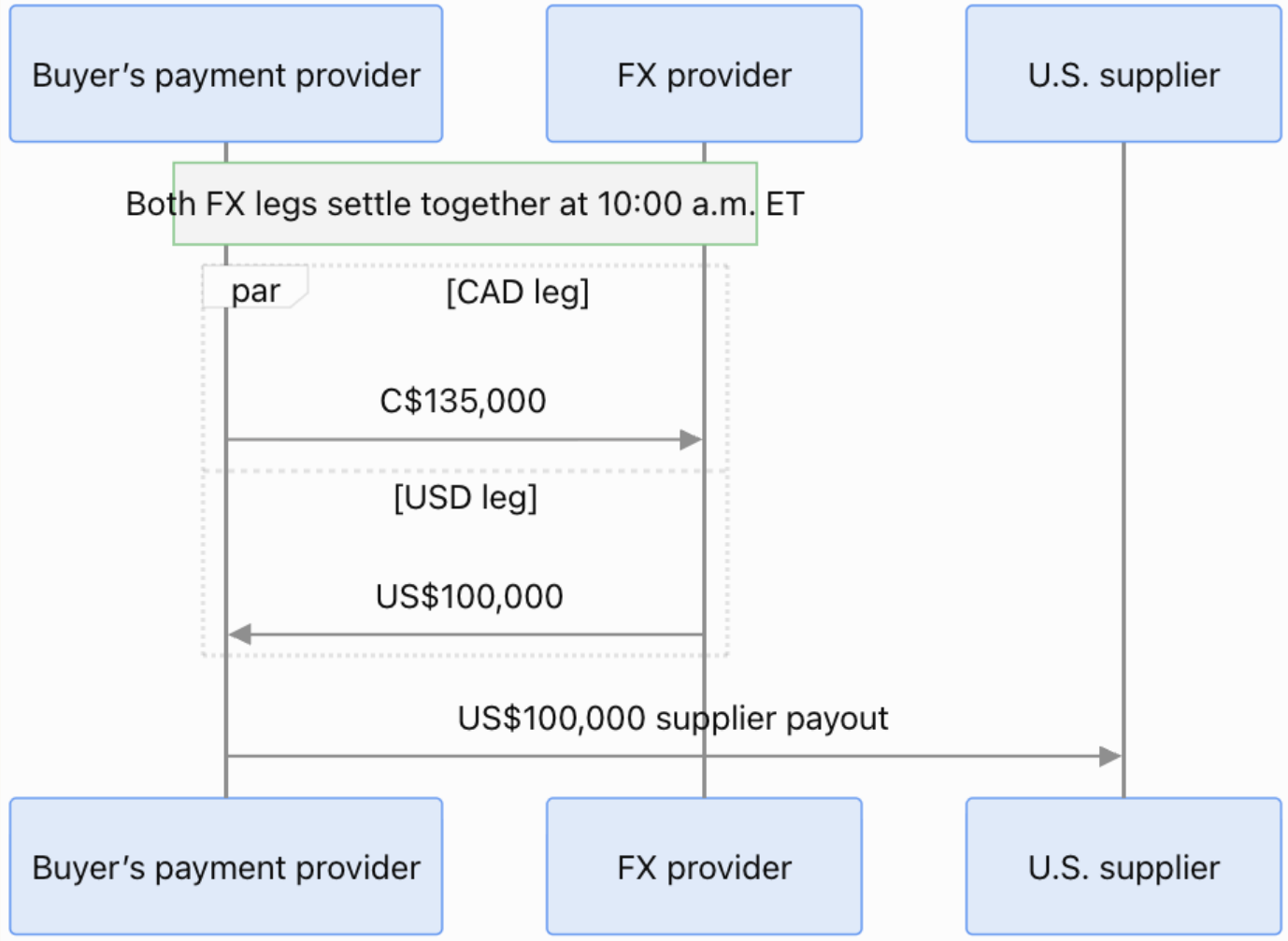
With tokenized deposits, a shared rule can trigger payout when both FX legs are final.

Illustrative FX rate of C$1.35 per US$1.

BIS · Project Agorá prototype

From invoice due time to spendable USD

Illustrative tokenized-deposit route for the US$100,000 invoice.



10:00 a.m. ET

The invoice is due and both FX legs settle together.

10:02 a.m. ET

The supplier can spend the full US$100,000 payout.

T = 2 minutes

Due time to spendable USD

A shorter wait shrinks the supplier's buffer

Same US$100,000 invoice. The tokenized route triggers payout when both FX legs settle.

$$\frac{SB_{2\ min}}{SB_{30\ min}} = \sqrt{(2/30)} \approx 0.258$$

74%

lower modeled supplier cash buffer

Illustrative comparison at equal peak funding	Synchronized instant rails	Tokenized deposits
Due time to spendable USD	30 min · 10:30 a.m. ET	2 min · 10:02 a.m. ET
Modeled supplier cash buffer[1]	US$10,000	**≈ US$2,580**
Peak cash reserved by buyer and FX provider[2]	C$270,000	C$270,000

[1] Same z and σ. Assumes independent minute-level forecast errors with constant variance.

[2] Both routes reserve C$135,000 from the buyer and US$100,000 from the FX provider.
With one hour of prefunding, the C$297,000 case uses 301,500 CAD-hours versus 337,500 for the 30-minute route.

Can a two-minute payout meet the same reliability target at the same funding limit?